

Article

FRFP - Big Picture

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1. Introduction

Large language models (LLMs) are increasingly used to perform analytical reasoning tasks across domains such as engineering, finance, governance, and scientific investigation. Despite rapid progress in model capabilities, a fundamental challenge remains: ensuring that analytical outputs produced by LLM-based systems maintain clear separation between machine-generated representations and human-grounded judgments.

The Foundational Reasoning and Feedback Protocol (FRFP) provides a structured framework for addressing this challenge. FRFP introduces a formal separation between *explicit representations*, which are machine-manipulable artifacts, and *tacit states*, which represent human competence, grounding, and endorsement. The protocol constrains reasoning systems so that machine processes operate strictly within explicit representation spaces while final acceptance or endorsement remains a human function.

The FRFP framework is developed through a multi-layer research program that spans theoretical foundations, formal validation, engineering translation, and empirical evaluation. The mathematical foundations of the framework are established in *FRFP Paper Vol. 1*, which defines the explicit–tacit separation, the representational boundary, and the formal constraints governing reasoning protocols. A companion tutorial provides a pedagogical explanation of the underlying mathematics, while a formalization in the *LEAN* theorem prover enables machine-checked validation of the theoretical structure.

Building on these foundations, *FRFP Paper Vol. 2* develops a corpus of canonical case studies involving complex socio-technical failures. These case studies demonstrate how the FRFP framework can be applied to analyze real-world systems involving interactions between technical mechanisms, organizational processes, and governance structures.

The theoretical framework is translated into implementable procedures through the *FRFP Technical Specification*. This specification converts the abstract protocol constraints into operational rules suitable for LLM-based reasoning pipelines, including explicit artifact discipline, navigation integrity, assumption quarantine, and separation between generation and endorsement.

Engineering-level descriptions of the protocol and its implications for AI reasoning systems are further developed through a series of FRFP white papers. These documents communicate the practical consequences of structured reasoning protocols for system design and illustrate how FRFP differs from traditional prompt-engineering approaches.

To empirically evaluate structured reasoning interventions, a general experimental framework is introduced in *Experimental Methods for Evaluating AI Agents*. This framework defines controlled procedures for constructing frozen analytical tasks, generating comparable model outputs under different reasoning conditions, and performing blinded human evaluation of analytical artifacts.

Within this framework, the *FRFP vs Base Survey* provides an empirical evaluation of protocol-guided reasoning. In this study, a set of historical socio-technical cases is analyzed under three conditions: baseline LLM analysis, protocol-guided LLM reasoning, and reconstructions derived from independent assessor reports. Model outputs are normalized into comparable artifacts and evaluated through blinded human comparison.

The survey relies on standardized one-page analytical summaries that present each model-generated analysis in a consistent format. These artifacts serve as the evaluation interface for human raters and allow direct comparison of reasoning outputs across experimental conditions.

Figure 1 summarizes the overall architecture of the FRFP research and validation program, illustrating how the theoretical, engineering, and empirical components interact.

Together, these components establish a complete research pipeline for developing, implementing, and empirically evaluating structured reasoning protocols for AI-assisted analytical systems.

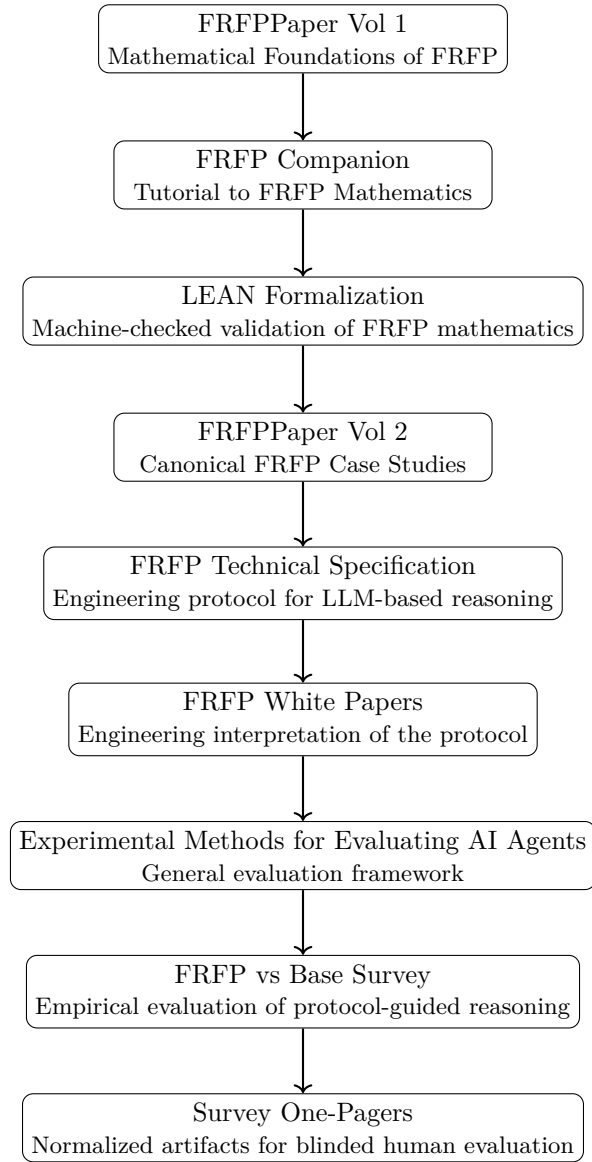


Figure 1. FRFP research and validation stack. Mathematical foundations are translated into engineering protocols and evaluated through controlled experimental methodology and empirical survey analysis.

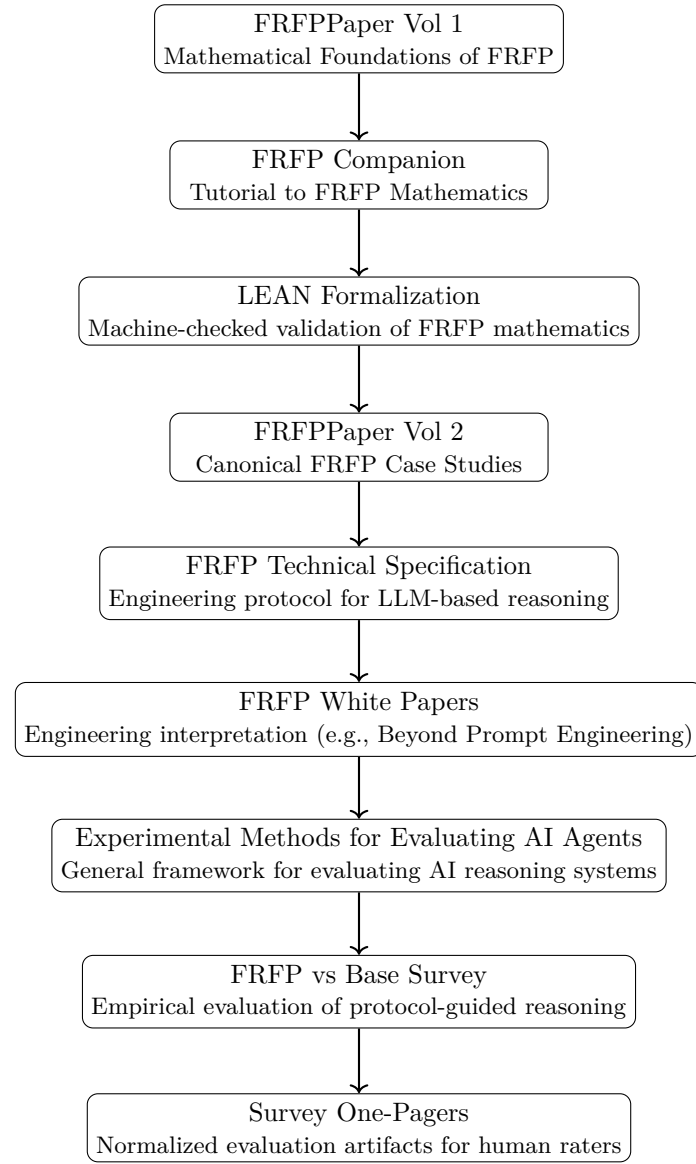


Figure 2. FRFP research and validation stack. Mathematical theory (top) is translated into engineering protocols and evaluated through controlled experimental methodology and empirical survey analysis.

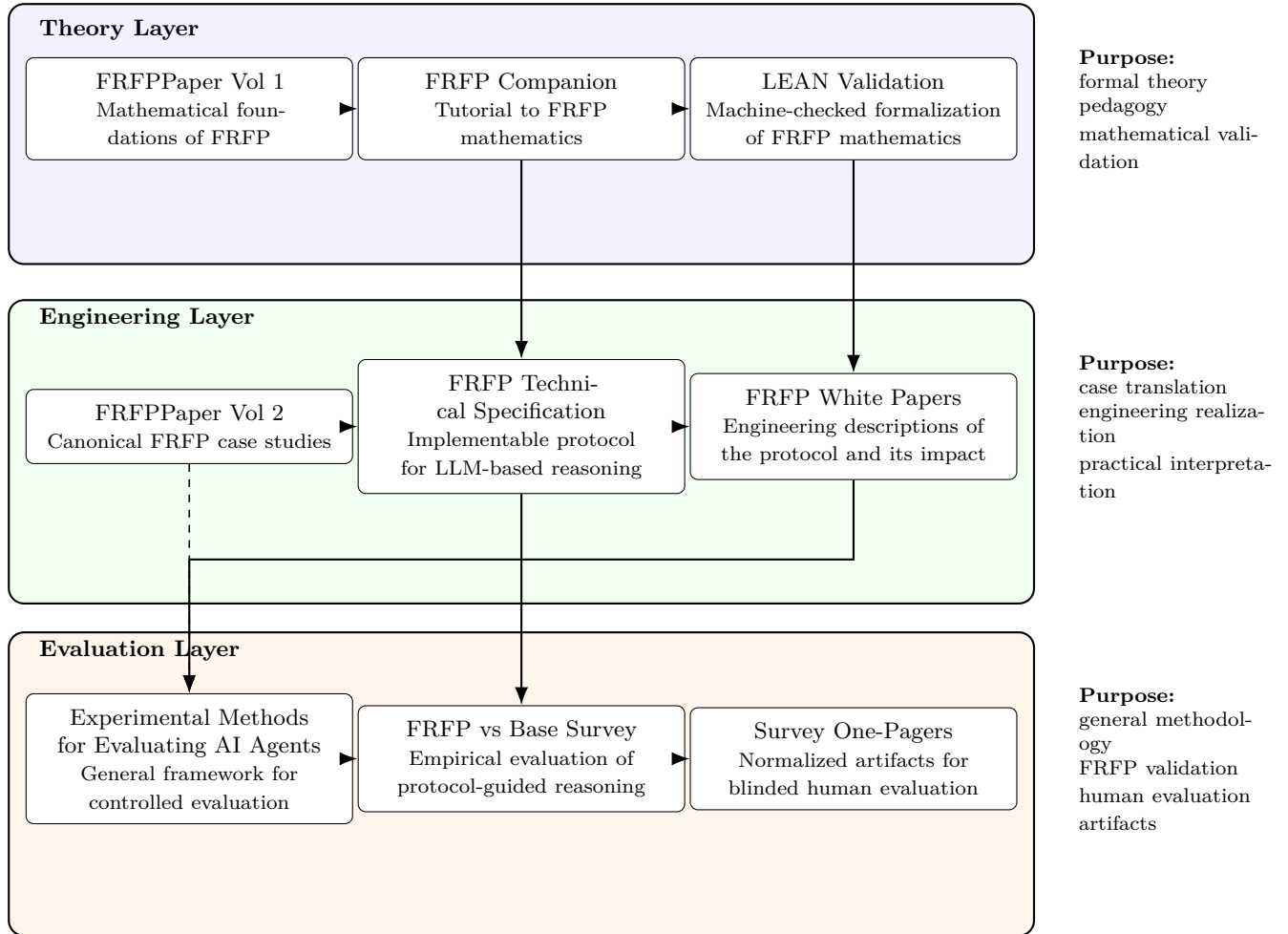


Figure 3. Three-layer FRFP architecture. The theory layer defines the formal foundations of the framework and supports pedagogical and machine-checked validation. The engineering layer translates the theory into canonical case studies, implementable protocol specifications, and engineering-facing white papers. The evaluation layer provides a general experimental framework, an FRFP-specific empirical survey, and normalized one-page artifacts for blinded human assessment.